Five Raters Are Not a Distribution: What Learned Judges Can Learn from Human Disagreement, and Where to Spend the Annotation Budget

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October 26, 2026

Abstract

Learned judges are increasingly trained on the full histogram of human ratings rather than on a single label, with the stated aim of learning how people disagree. We ask what part of such a histogram a text-conditioned judge can learn at all. We separate two properties that prior work conflates: whether the histogram's shape beyond its location is *reproducible across raters*, and whether it is *predictable from the text*. Across 21 annotation panels (summarisation, offensiveness, politeness, helpfulness, sarcasm, valence, conversational safety, NLI, and five software-engineering panels including a 5,468-developer code-readability study), a learner-free split-half analysis shows that with 3–9 raters the shape beyond mode and mean is net noise for forecasting a new rater, and that dense ordinal panels need 20–30 raters before the raw histogram beats a mode-and-mean comparator. A cross-fitted decoding analysis shows that the reproducible part of the shape is not linearly decodable from the frozen representations of three model families (Qwen2.5-7B, Qwen3-8B, OLMo-2-7B) on any of 13 ordinal panels ($|G| \leq 6{\cdot}10^{-5}$, final or best layers), but is on NLI ($G = +0.003$ to $+0.004$ in every family). Learned judges behave accordingly: statistic-preserving replacement of the shape of 70–80 % of training targets leaves LoRA judges unchanged, and no construction of the per-item target moves frozen-feature heads by more than ± 0.0005 RPS on 11 panels. An ordered-probit decomposition attributes the ordinal structure beyond location to rater thresholds — a property of the people, not the text. The practical consequence is a budget rule, validated end-to-end: at a fixed number of ratings, judges improve with more items rather than more raters per item (on SummEval, 960 items $\times$ 1 rating beats 960 $\times$ 7); the marginal value of further ratings per item is task-dependent. We release per-dataset disagreement reliability cards and the estimators, and conclude that raters are for evaluation sets, items for training sets.

1 Introduction

Human raters disagree, and a growing line of work trains learned judges — language models fine-tuned to score text — on the full distribution of human ratings rather than on a single aggregated label (Chen et al., 2025; Singh et al., 2025; Kohli, 2026; Benito-Santos and Ghajari, 2025). The appeal is clear: a histogram records which neighbouring levels received mass and how concentrated the ratings were, and a judge that reproduces it would "capture human disagreement" rather than average it away. The data behind that appeal are thin. In most datasets the histogram is computed from three to nine people. Five ratings of a summary's coherence are a sample from a population of possible raters, and a model trained to reproduce the sample exactly is, in part, trained to reproduce sampling noise.

This paper asks a question that the literature on learning from label variation (Uma et al., 2021; Plank, 2022) has not asked in this form: **which part of a few-rater histogram can a text-conditioned judge learn?** We decompose every histogram into its *location* (mode and mean) and the *shape beyond location* (the residual allocation of mass once mode and mean are fixed), and we measure two properties of the shape that are usually conflated:

1. **Reproducibility across raters.** If the same items were rated by a disjoint panel, would the shape come out the same? We estimate this learner-free with a split-half design: an exact identity, $A1 = M - 2X$, relates the forecasting value of the raw histogram to the magnitude M of its residual and the cross-panel agreement X of that residual; the ratio $\rho = X/M$ is a split-half reliability of the shape, and the raw histogram beats a mode-and-mean comparator for predicting a new rater exactly when $\rho > \frac{1}{2}$.

2. **Predictability from the text.** Even a reproducible shape is useless to a judge unless the text carries it. We fit a cross-fitted ridge decoder from a frozen LLM's representation to the panel-1 residual and score it on an independent panel-2 residual with a proper-score gain G, against a within-cell feature-permutation control and a positive control.

Reliable is not the same as learnable. The two measurements come apart in a way that organises everything else we find.

Findings. (i) With 3–9 raters, the shape beyond mode and mean is net noise for forecasting a new rater on every ordinal panel we study (A1x > 0 on all eight small panels; with three raters there is no shape at all — on the code-summarisation benchmark of Crupi et al. (2025) and on MLCQ the mode-and-mean cell determines the histogram). Dense ordinal panels — Buse–Weimer, DICES, and the three languages of Dorn's 5,468-developer readability study — need 20–30 raters before the raw histogram beats the comparator; ChaosNLI needs 5–10. (ii) The reproducible part of the shape is not linearly decodable from the frozen representation of the same 7B model that the judges are built on: $|\mathrm{G}| \leq 6{\cdot}10^{-5}$ on 13 ordinal and software-engineering panels, whether we probe the final layer or greedily select the best layers, while the positive controls (predicting the item's mean) reach Spearman 0.3–0.7. On the two ChaosNLI panels, G $= +0.0032$ and $+0.0043$ with the permutation control negative: NLI disagreement is in the text; quality-rating disagreement is not. (iii) Learned judges behave as these measurements predict. Replacing the shape of 70–80 % of training targets while preserving each target's mode and mean leaves LoRA judges unchanged on SummEval (-0.0001 [-0.0011, $+0.0011$]) and POPQUORN ($+0.0000$ [-0.0009, $+0.0009$]), whereas the same intervention on ChaosNLI costs 0.016–0.052 KL; and no construction of the per-item training target — shrinking its shape or its location, Dirichlet smoothing, label smoothing toward uniform or toward the marginal — moves frozen-feature heads by more than ± 0.0005 RPS on 11 panels, while discarding the distribution costs 0.003–0.010. (iv) An ordered-probit decomposition on the panels with persistent rater identities attributes the ordinal structure beyond location to **rater thresholds**: allowing each rater their own cut-points improves held-out likelihood by 0.05–0.16 nats per rating, item-specific dispersion adds at most 0.012, and on the dense panels no reproducible item-specific shape beyond mode and mean remains. Disagreement on a rating scale is mostly about who is rating, which is why the text cannot carry it. (v) The budget consequence holds end-to-end. At a fixed number of ratings, frozen-feature heads improve with more items rather than more raters on 11 of 11 panels with $\gtrsim 300$ items, and LoRA judges do too: on SummEval, 960 items $\times$ 1 rating (960 ratings) yields a lower test RPS than 960 $\times$ 7 (6,720 ratings; -0.0061 [-0.0087, -0.0035]), and 144 $\times$ 7 is far worse ($+0.0211$); on POPQUORN the ordering is the same at both budgets, although there a second rating per item still buys about half the single-rating gap. A prediction we registered — that multi-rater items would pay off at fixed budget where the shape is text-predictable — was falsified on ChaosNLI (items first there too); the marginal value of additional ratings per item is where tasks differ, and we report it as an observation, not a claim.

Contributions. (1) Two reliabilities for disagreement — across raters and across text — with learner-free, cross-fitted estimators, and the empirical dissociation between them on ordinal versus nominal tasks. (2) Disagreement reliability cards for 21 panels, including five software-engineering panels, that tell a dataset builder how many raters a task needs before the shape of disagreement is a property of the item. (3) A rater–item decomposition that explains the ordinal result and bounds what any text-conditioned judge can learn without rater identity. (4) A budget rule — cover items before adding raters — validated with frozen-feature heads on 11 panels and with end-to-end LoRA judges on two. The donor-replacement audit of our earlier draft becomes the motivating evidence that current training does not use the shape; shrinkage of the histogram survives as a tool for building reference distributions, not as a training method. This paper is the training-side companion of our evaluation-side study (paper 1): evaluation should not average disagreement away; training mostly cannot learn it. Raters are for the test set, items for the training set.

What we do not claim. That disagreement is noise (its reproducible part is not); that the text contains no residual information (we test linear decodability from three model families); a theoretical rater-count threshold (a two-term $1/a$ model was falsified); or a causal role of label structure in the ordinal–nominal difference (we offer a measured mechanism, rater thresholds, and the decoding contrast).

2 Two reliabilities of a disagreement histogram

Setting and notation. Item i has ratings on K ordered levels from a panel of raters; the empirical histogram is h_i. For ordinal scales we score forecasts with the ranked probability score, $S(f, y) = \sum_k (F_f(k) - \mathbf{1}[y \leq$

$k])^2/(K-1)$, and write C for the cumulative operator; for nominal tasks (ChaosNLI) S is the Brier score and C the identity. The **location** of a histogram is its exact mode and mean (nominal: modal class and modal mass); the **shape beyond location** is what remains once these are fixed. To isolate it we use the *exact-T comparator* q_T of our earlier audit: a cross-fitted forecast built from other items in the same exact mode-and-mean cell (KL-projected onto the cell when the cell is sparse), so that q_T has the same mode and mean as h_i and differs from it only in shape. The residual of a histogram is $r = C(h - q_T)$.

2.1 Reproducibility across raters (split-half reliability).

In each of R repetitions the ratings of every item are split at random into disjoint panels P_1 (size a) and P_2. Writing $r_1 = C(h_1 - q_T)$ and $r_2 = C(h_2 - q_T)$, with q_T computed from P_1, the forecasting value of the raw histogram over the comparator obeys the exact identity

$$A1 \;=\; \mathbb{E}\big[S(h_1, y) - S(q_T, y)\big]_{y \in P_2} \;=\; M - 2X, \qquad M = \|r_1\|^2/n,\; X = \langle r_1, r_2\rangle/n,$$

so $\rho = X/M$ is a split-half reliability of the shape and the raw histogram beats the location-only forecast exactly when $\rho > \frac{1}{2}$. M is the observed magnitude of the shape, X its agreement with an independent panel's shape; $M - X$ is sampling noise. We also report $A1$ with a smoother, response-based comparator that does not reproduce the noisy location (the primary estimand of our finite-rater protocol), and the support of the donor-replacement intervention (the fraction of items whose histogram could be changed at all while preserving mode and mean). Intervals are item-cluster bootstraps that refit the comparator in every replicate. Table 1 gives the resulting *reliability cards* for 21 panels; Figure 1 traces ρ and $A1$ against the number of raters a on the dense panels.

2.2 Predictability from the text (decodability).

A judge sees the text, not the raters. Whether the reproducible shape is available to it is a separate question, which we answer with the representation the judges are built on: last-token hidden states of the frozen Qwen2.5-7B-Instruct (Qwen2.5-Coder for code panels) under the same rating prompt. A multi-output ridge decoder, group-cross-fitted over five item folds with a label-free PCA and a nested penalty, predicts r_1 from the features; it is scored on the *independent* panel r_2 by the covariance $V = \langle \hat{r}, r_2\rangle/n$ and by the proper-score gain

$$G \;=\; \big(2\langle \hat{r}, r_2\rangle - \|\hat{r}\|^2\big)/n \;=\; \mathbb{E}_{y \in P_2}\big[S(q_T, y) - S(q_T + C^{-1}\hat{r},\, y)\big],$$

the exact reduction in score obtained by adding the decoded shape to the location-only forecast. Two controls accompany every estimate: a *within-cell permutation* of the features among items with the same exact mode and mean (which must drive G to zero or below), and a *positive control* that predicts the panel-1 mean with the same pipeline (which must be clearly positive, else the representation or the alignment is broken). Directions are symmetrised ($P_1 \leftrightarrow P_2$) and bootstrapped over item clusters. Because a single layer is a weak probe, we repeat the analysis with per-layer mean- and last-token features and greedy selection of the best layers (the probing recipe of Li et al., 2026).

2.3 Why the two can come apart.

ρ asks whether the shape is a property of the item's *rater population*; G asks whether it is a property of the item's *text*. A shape can be reproducible yet invisible to text — for instance when it is produced by how particular kinds of raters use the scale — and a text-predictable shape can be unreliable with few raters. Section 4 shows that the first case is the ordinal norm. Nothing in §2.1 or §2.2 trains a judge; both are measurements of the data and the representation, and both are released with the reliability cards so that a dataset builder can compute them before deciding how many raters to pay for.

3 What learned judges use: the audit, condensed

If a judge learned the shape beyond location, destroying that shape in its training targets would hurt it. Our donor-replacement audit (Sections 2–4 of the previous version; Tables T2b, T3b, T3c, T5) does exactly this with certified dose: for 70–83 % of training items the histogram is replaced by another item's histogram projected

onto the recipient's exact mode and mean (mean TV 0.30–0.35 among changed items), under two learners. **End-to-end LoRA judges** (Qwen2.5-7B-Instruct, the distributional objective of Chen et al., 2025; three optimizer–assignment realisations × three held-out-rater folds) change by −0.0001 [−0.0011, +0.0011] RPS on SummEval coherence and +0.0000 [−0.0009, +0.0009] on POPQUORN offensiveness, whereas destroying the location as well (mode-only replacement) costs +0.0120 [+0.0086, +0.0155] and +0.0029 [+0.0009, +0.0050] with every seed-by-fold cell positive. **Frozen-feature heads** on ten ordinal panels move by −0.0024 to +0.0006. On the nominal ChaosNLI panels the same kind of intervention — scrambling the allocation of mass beyond the modal class while preserving the modal mass — raises test KL by 0.016–0.052 at every annotation count.

A stronger version of the same test asks whether *any* construction of the per-item target helps. We trained the heads on eleven ordinal panels with the raw histogram, the location-only comparator, reliability-shrunk shapes and locations, Dirichlet smoothing toward the mode prototype, label smoothing toward uniform and toward the marginal, and a cross-item smoothed comparator, thirty configurations each. No construction changes held-out RPS by more than ±0.0005 with a consistent sign across panels; the one-hot target costs +0.003 to +0.010 everywhere. The learner pools location across items on its own and is indifferent to the per-item shape — exactly what §2.2 predicts for a shape it cannot see. Learned judges on ordinal scales learn location and calibration; on NLI they also learn shape, because there the shape is in the text.

4 Where the ordinal shape comes from: a rater–item decomposition

The split-half analysis says the few-rater shape is mostly noise but not entirely (X > 0 on every panel); the decoding analysis says the reproducible part is not in the text. What is it, then? We fit three ordered-probit models to the rating matrices of the panels with persistent rater identities (Buse–Weimer, DICES-350, Scalabrino, the SummEval experts) and location-plus-dispersion models to the position panels: $Y^*_{ir} = \theta_i + \sigma_i\,\varepsilon_{ir}$ with cut-points c_r; **M0** item location with common cut-points, **M1** adding rater-specific cut-points (harshness), **M2** adding item-specific dispersion σ_i (the inverse of IRT discrimination). Held-out log-likelihood over random cells, five folds, with the dispersion penalty chosen on held-out data.

Rater thresholds carry the structure: M1 improves held-out likelihood by +0.111 (BW), +0.146 (DICES-350), +0.156 (Scalabrino), +0.104 / +0.047 (SummEval coherence / relevance) nats per rating; item dispersion adds +0.001 to +0.012 and over-fits on every 5–8-rating panel. Simulating rating matrices from the fitted models and re-running the split-half analysis closes the loop: on the dense ordinal panels the observed exact-T residual reliability is X ≈ 0 (BW −0.0001, DICES-350 +0.00003, DICES-990 +0.00002) and every model reproduces it — once mode and mean are fixed there is **no reproducible item-specific shape** on these panels at all. SummEval retains a small residual (X = +0.0018 / +0.0016) that neither thresholds nor one-dimensional dispersion reproduce, and it is precisely this residual that the decoder of §2.2 cannot read from the text. The mechanism is the textbook one of many-facet Rasch measurement (Linacre, 1989; Eckes, 2011), and we present it as measurement, not as a claim about raters' psychology; its consequence for judges is new: on ordinal rating scales the disagreement a text-conditioned model is asked to learn is largely a property of the people who rated, which no amount of text can supply without rater identity.

The annotation side confirms the same picture from a different angle. For forecasting a new rater from an item's own ratings, shrinking the *location* toward the mode prototype by its split-half reliability improves RPS by −0.0048 [−0.0051, −0.0044] over the raw histogram (eight panels pooled), a cross-item smoothed comparator by −0.0113; shrinking the *shape* by its reliability $\lambda = X/M$ beats the raw histogram (−0.0021) but is indistinguishable from the location-only comparator (−0.0000 [−0.0000, −0.0000]) on every panel except SummEval, where $\lambda \approx 0.4$. What a few-rater histogram contributes beyond its location is, for practical purposes, nothing.

5 Spending the annotation budget

If the shape is unlearnable and the location is what the judge learns, then every rating after the first on the same item buys at most a less noisy location, while a rating on a new item buys a new point for the regression the judge is implicitly fitting. We tested this directly by holding the total number of ratings fixed and trading items for raters.

Frozen-feature heads (nine ordinal panels, thirty configurations each; Table 5). On every panel with more than about 300 training items, more items with fewer ratings each wins monotonically: on DICES-990 at

a budget of 594 ratings, 594 items × 1 rating gives RPS 0.203, 297 × 2 0.205, 148 × 4 0.207, 74 × 8 0.213 and 40 × 15 0.219; POPQUORN, HelpSteer2 and EmoBank follow the same order, SummEval is flat. The direction reverses only on small item sets (Buse–Weimer with 60 training items, DICES-350 with 210), where a single rating per item is too noisy a target. The same items-first order holds on ChaosNLI (959 × 1: KL 0.226; 30 × 32: 0.323) — a result that falsified a prediction we had registered, namely that multi-rater items would pay off at fixed budget where the shape is text-predictable. They do not; where tasks differ is the *marginal* value of a second rating on already-covered items (ChaosNLI −15 %, ordinal heads −2 to −3 %), which we report as an observation.

End-to-end LoRA judges (Table 6). On SummEval coherence, with the Chen et al. objective and the three held-out-expert folds of the audit, a judge trained on 960 items × 1 randomly chosen rating (960 ratings) reaches a lower test RPS than the judge trained on all 960 × 7 = 6,720 ratings (−0.0061 [−0.0087, −0.0035], crossed bootstrap over seeds and test documents); 480 × 2 is equally good (−0.0052), 240 × 4 is worse (+0.0162) and 144 × 7 much worse (+0.0211); adding a second rating to every item (960 × 2) changes nothing (−0.0064). On POPQUORN offensiveness the ordering within each budget is again items-first (819 × 1 best at 820 ratings, 819 × 2 best at 1,638), but here extra ratings per item do help: 819 × 1 is +0.0077 [+0.0046, +0.0110] worse than the full 5,733-rating data and 819 × 2 recovers two thirds of that gap. Single-rating training behaves like stochastic label sampling (Gheibi and Ghazizadeh, 2026) and is consistent with the many-once-over-few-many results for hard-label classifiers (Khetan et al., 2018; Lin et al., 2014); our contribution is the demonstration with distributional targets and end-to-end LLM judges, and the explanation of §§2–4 for why the raters' side of the budget buys so little.

Rule. For training a judge on an ordinal rating task, cover the items first; a second rating per item is worth between nothing (SummEval) and about half of the single-rating gap (POPQUORN), and further ratings less. Raters belong where a reliable reference distribution is itself the deliverable — evaluation sets (our companion paper), and tasks such as NLI where the shape is in the text.

6 Software-engineering panels

Readability, summary quality and code-smell severity are rated on the same few-rater scales, and the same questions arise for code judges. Dorn's readability study (360 snippets in Java, Python and CUDA; 161–267 developers per snippet) gives the densest ordinal panel we know of: its reliability curve matches Buse–Weimer and DICES on every language — ρ rises from 0.05 at two raters to 0.6 at forty, and the raw histogram does not beat the mode-and-mean comparator before 20–30 raters (Figure 1). The LLM-judge validation benchmarks of the field have three raters per item — Crupi et al.'s code summaries, MLCQ's smell severities, Roy et al.'s summaries — and with three raters the mode-and-mean cell determines the histogram: there is no shape to learn or to evaluate (support 0 %, A1x ≈ 0). Scalabrino's nine-rater panel and JetBrains' 11–31-developer panel sit in between. The decoder reads no shape from the Coder model's representation on any of them (Dorn $|G| = 2{\cdot}10^{-5}$, Scalabrino, JetBrains ≈ 0). For SE human studies this yields two concrete numbers: three to five raters per artifact fix its location and nothing else; evaluation sets that are meant to measure disagreement need 20–30.

7 Limitations

Base models: all frozen-feature results (decoding, target construction, fixed budget) replicate on Qwen3-8B and on the non-Qwen OLMo-2-7B (reports/CROSS_FAMILY_DECODING.md: ordinal $|G| \leq 1{\cdot}10^{-5}$ on 6 panels in each family, ChaosNLI +0.0033/+0.0034; budget orderings within 0.002 RPS of each other; target constructions within ±0.0005 except location-shrinking ones), but the end-to-end LoRA results use Qwen2.5-7B only; frozen-feature heads for breadth, end-to-end judges on two datasets; linear decoders — we re-tested with per-layer features and greedy layer selection, not with non-linear or fine-tuned probes; SummEval's small unexplained residual; POPQUORN's position folds (rater identities are available and a user-held-out replication is pending); one registered prediction falsified and reported; effect sizes in the third decimal of RPS, to be read against the mode-only penalty of +0.012 and the budget effects of +0.02. We make no claim that disagreement is noise, that text contains no residual information, that a theoretical rater count exists, or that label structure causes the ordinal–nominal difference.

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